

Mohamed Amine **AROUS**

AI Engineer · LLMs, RAG, AI Agents & Knowledge Graphs

ENSTA Paris graduate · Open to relocation

CONTACT

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destivano.github.io
github.com/Destivano

LinkedIn ↗
Google Scholar ↗

LANGUAGES

Arabic · Native

English · C1, Linguaskill

French · C1, TCF

TECHNOLOGIES

Python · PyTorch · scikit-learn · Neo4j · FastAPI · Docker · PostgreSQL · NumPy · pandas · Git · TensorFlow · GitHub

Generative AI & Retrieval
LLMs, RAG/GraphRAG, AI agents, Transformers, text embeddings, BM25, CrewAI, Ollama.

Graphs & Data
OWL, RDF, SHACL, Neo4j, PostgreSQL, DynamoDB, SQL.

Evaluation & Engineering
Automated testing, model and retrieval evaluation, REST APIs, Git, Docker, CI/CD, GPUs, Slurm.

Other Languages
C++, SQL.

AI engineer trained at ENSTA Paris, with research experience at CEA List and École Polytechnique. I develop and evaluate systems combining language models, knowledge graphs and information retrieval, from prototyping to deployment.

Chat with my AI assistant ↗
Explore my background and projects through answers backed by sources.

EXPERIENCE

Apr – Oct 2026

CEA List · LASTI Team

Research Intern · Palaiseau, France

- Built a Python pipeline enriching 3D scene graphs with expert rules, Wikidata and LLM proposals; applied consistency, plausibility and safety checks.
- Improved retrieval for warehouse robotic reasoning: *Hit@1 from 21.01% to 31.96% (+10.95 percentage points)*; safety constraints blocked *250+ navigation tasks* in benchmark tests.

Supervisor: Gaël de Chalendar · LinkedIn recommendation ↗

Renault · project collaborator

May – Sep 2025

École Polytechnique · CMAP

Research Intern · Palaiseau, France

- Adapted a speech synthesis model for French and German using Python/PyTorch; contributed to the open-source CosyVoice2-EU project.
- Compared models and contributed to an *EUSIPCO 2026* paper: lowest word error rate among the open-source models compared, on data outside the training domain.

Supervisor: Prof. Eric Moulines

France 2030 · associated programme

EDUCATION

2024–2026

ENSTA Paris — Engineering Degree (Diplôme d'ingénieur)

ENSTA Paris–ENIT double degree, Master's level — *Palaiseau, France*

AI specialisation · 2nd-year GPA: *3.96/4.3*.
Deep Learning, Machine Learning, Language Modeling, Reinforcement Learning, game theory and multi-agent control, probabilistic reasoning.

2023–2024

ENIT — Engineering Studies in Telecommunications

ENSTA Paris–ENIT double-degree programme — *Tunis, Tunisia*

1st-year rank: *2/50*.

2021–2023

IPEIT — Preparatory Classes in Mathematics and Physics

Institut Préparatoire aux Études d'Ingénieurs de Tunis — *Tunis, Tunisia*

Rankings: *4/177* (2nd year), *3/326* (1st year).
Linear algebra, probability, statistics, numerical analysis, physics, chemistry, Python and SQL.

From ideas to **applications**

Personal projects, experiments and engineering

> BUILD_

CERTIFICATIONS

AWS Certified AI Practitioner
Amazon Web Services · 2026
Verify certificate ↗

Exam preparation in progress
AWS Certified Cloud Practitioner
AWS Certified Solutions Architect – Associate

Generative AI Engineering Professional Certificate
IBM · 2025 · 16 courseses
Verify certificate ↗

Data Science Professional Certificate
IBM · 2025 · 6 courses
Verify certificate ↗

Foundations of AI Professional Certificate
IBM · 2025 · 3 courses
Verify certificate ↗

AWARDS

Google MENA ML 2026
Full scholarship · Google DeepMind

Excellence scholarship
Tunisian Government

School ambassador · 5 times
Competitive programming and AI

CLOUD & DELIVERY

AWS Bedrock, Lambda, SQS, DynamoDB, CloudWatch, EC2, IAM, CodePipeline, CodeBuild, CodeDeploy, CodeArtifact.

SELECTED PROJECTS

OpsPilot — AI Agent with Tools and Human Approval
Nov 2026

Prototype · simulated business operations GitHub ↗ Demo ↗

- Python/AWS Bedrock agent orchestrating simulated tools, with persistent DynamoDB state, human approvals and CloudWatch execution traces.
- Tested *30 simulated scenarios*. Across the *22 cases in the live-model evaluation protocol*, fixes increased successful outcomes from *12 to 19*.

Portfolio — RAG-Powered AI Assistant
2026

Deployed personal application Frontend ↗ Backend ↗ Demo ↗

- FastAPI on AWS Lambda with hybrid BM25/embedding retrieval over my career information.
- *413 passing tests*; no hallucinations observed on the *18-question* evaluation set.

NORMA — GraphRAG for Automotive Regulations
2026

Bilingual, versioned and traceable knowledge graph GitHub ↗ Demo ↗

- Structured UNECE/EU texts in Neo4j using an OWL ontology and SHACL validation: *21,700 violations reduced to 0*.
- Used clause relationships for multi-hop retrieval: *Recall@10 from 0.317 to 0.708* on the internal benchmark through deterministic link expansion.

AWS CI/CD Pipeline
2026

Automated builds and deployment GitHub ↗

- Deployed a web application from GitHub to EC2; automated builds, artifacts and releases using CodePipeline, CodeBuild, CodeDeploy, CodeArtifact and IAM.

Embodied AI — VLA Agent Benchmark
Oct 2025 – Mar 2026

Academic project · team of 5 · Prof. Gianni Franchi GitHub ↗

- Built an automated benchmark comparing Qwen3-8B and Llama3-8B within the same AIUTA agent, running evaluations on a GPU cluster.

STMicroelectronics — Component Similarity
Oct 2024 – May 2025

Academic project with an industrial partner

- Developed machine learning models in Python (scikit-learn, PyTorch, pandas, NumPy) to evaluate similarity between microelectronic components.
- Designed and validated a solution within a multidisciplinary engineering team, achieving *98% accuracy* and outperforming the company's internal baseline.

VOLUNTEERING & LEADERSHIP

Tutor — #GENIUS Programme, École Polytechnique
Weekly tutoring in science and mathematics.

Competitive Programming Lead — G²FOSS
Organised data structures and algorithms courses and programming contests.

PUBLICATION

Europeanizing Modular Zero-Shot TTS: A Component-Level Adaptation Framework for French and German.
EUSIPCO 2026 · Paper ↗ · Code ↗